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Statistical inference in vector autoregressions with possibly integrated processes

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Abstract

This paper shows how we can estimate VAR's formulated in levels and test general restrictions on the parameter matrices even if the processes may be integrated or cointegrated of an arbitrary order. We can apply a usual lag selection procedure to a possibly integrated or cointegrated VAR since the standard asymptotic theory is valid (as far as the order of integration of the process does not exceed the true lag length of the model). Having determined a lag length k , we then estimate a $(k + d_{\max})$ th-order VAR where $d_{\max}$ is the maximal order of integration that we suspect might occur in the process. The coefficient matrices of the last $d_{\max}$ lagged vectors in the model are ignored (since these are regarded as zeros), and we can test linear or nonlinear restrictions on the first k coefficient matrices using the standard asymptotic theory.

Key words: Cointegration; Hypothesis testing; Lag order selection; Unit roots; Vector autoregressions

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1. Introduction

Vector autoregressions (VAR's) are one of the most heavily used classes of models in applied econometrics. However, Park and Phillips (1989) and Sims, Stock, and Watson (1990) among others have recently shown that the

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conventional asymptotic theory is, in general, not applicable to hypothesis testing in levels VAR's if the variables are integrated or cointegrated. If economic variables were known to be, say, $I(1)$ (integrated of order 1) with no cointegration, then one could estimate a VAR in first-order differences of the variables so that the conventional asymptotic theory is valid for hypothesis testing in the VAR. Similarly, if the variables were known to be, for example, $CI(1, 1)$ (cointegrated of order 1, 1), then one would specify an error correction model (ECM). But, in most applications, it is not known *a priori* whether the variables are integrated, cointegrated, or (trend) stationary. Consequently, pretests for a unit root(s) and cointegration in the economic time series (and estimation of the cointegrating vector(s) if there is cointegration) are usually required before estimating the VAR model in which statistical inferences are conducted.

Several tests for a unit root(s) in a single time series are available (e.g., Dickey and Fuller, 1979; Fuller, 1976; Pantula, 1989; Phillips, 1987; Phillips and Perron, 1988). Unfortunately, however, the power of these tests are known to be very low against the alternative hypothesis of (trend) stationarity. Tests for cointegration and cointegrating ranks have also been developed (e.g., Johansen, 1988, 1991; Phillips and Ouliaris, 1990; Stock and Watson, 1989). In particular, Johansen's method is related to the topic of the present paper since it is based on a VAR representation of the time series. Again, however, simulation experiments show that the tests for cointegrating ranks in Johansen-type ECM's are very sensitive to the values of the nuisance parameters in finite samples and hence not very reliable for sample sizes that are typical for economic time series (e.g., Reimers, 1992; Toda, 1995). These observations imply that the usual strategy that one tests some economic hypothesis *conditioned* on the estimation of a unit root, a cointegrating rank, and a cointegrating vector(s) may suffer from severe pretest biases.

Of course, this kind of problems is something econometricians have to live with if their interests are in the cointegrating relations themselves. In many applications of VAR models, however, the researcher's interest is not in the existence of unit roots or cointegrating relations themselves, but rather in testing economic hypotheses expressed as restrictions on the coefficients of the model. If that is the case, it is clearly desirable to have a testing procedure which is robust to the integration and cointegration properties of the process so as to avoid the possible pretest biases.

A typical example is the test of Granger causality in the VAR framework, where the null hypothesis is formulated as zero restrictions on the coefficients of the lags of a subset of the variables. As Sims, Stock, and Watson (1990, Example 2, Sect. 6) and Toda and Phillips (1993a, Sect. 3) show, the usual Wald test statistic for Granger noncausality based on levels estimation not only has a nonstandard asymptotic distribution but depends on nuisance parameters in general if the process is $I(1)$. Mosconi and Giannini (1992) and Toda and

Phillips (1993a, Sect. 4) applied Johansen's (1988, 1991) ECM estimation to the problem of Granger causality tests in $I(1)$ systems. The former is based on the likelihood ratio (LR) principle and the latter on the Wald principle, but both test procedures require the pretests of cointegrating ranks and those procedures are not very simple to implement. Moreover, a difficulty arises in these approaches since the noncausality hypothesis in ECM's involves nonlinear restrictions on parameter matrices, and therefore Wald (and presumably LR) tests for Granger noncausality may suffer from size distortions due to rank deficiency that cannot be excluded under the null hypothesis (see Toda and Phillips, 1993a).

The present paper proposes a simple way to overcome the problems in hypothesis testing that we encounter when VAR processes may have some unit roots. Our method is applicable whether the VAR's may be stationary (around a deterministic trend), integrated of an arbitrary order, or cointegrated of an arbitrary order. Consequently, one can test linear or nonlinear restrictions on the coefficients by estimating a levels VAR and applying the Wald criterion, paying little attention to the integration and cointegration properties of the time series data in hand.

The organization of the paper is as follows. Section 2 deals with the general model and provides an intuitive discussion on why our approach guarantees the validity of the conventional asymptotic theory in hypothesis testing based on levels VAR's even if the processes are nonstationary. In Section 3, for simplicity, we restrict our attention to the model where the variables are *at most* $I(2)$, and prove formally the above mentioned result, viz., one can test general restrictions on the parameters of levels VAR's using the conventional asymptotic theory. In Section 4, we consider the problem of choosing lag lengths of VAR's with possibly integrated or cointegrated processes. Concluding remarks are made in Section 5, and proofs of the lemmas used in the body of the paper are given in the Appendix.

A summary word on notation. We use $\text{vec}(M)$ to stack the *rows* of a matrix M into a column vector. $I(d)$ and $CI(d, b)$ denote an integrated process of order d and a cointegrated process of order d, b , respectively. We use the symbols ' $\xrightarrow{p}$ ', ' $\xrightarrow{d}$ ', and ' $\equiv$ ' to signify convergence in probability, convergence in distribution, and equality in distribution, respectively. The inequality ' > 0 ' denotes positive definite when applied to matrices. $[x]$ signifies the integer part of a real number x . All limits given in this paper are taken as the sample size $T \rightarrow \infty$.

2. The general model

Let an n -vector time series $\{y_t\}_{t=-k+1}^{\infty}$ be generated by the following model:

$$y_t = \beta_0 + \beta_1 t + \cdots + \beta_q t^q + \eta_t, \quad (1)$$

where $\{\eta_t\}$ is $I(d)$ and may be $CI(d, b)$. In particular, we assume that $\{\eta_t\}$ is a k th-order vector autoregressive process,

$$\eta_t = J_1 \eta_{t-1} + \cdots + J_k \eta_{t-k} + \varepsilon_t, \quad (2)$$

where k is assumed to be known¹ and

(A1) $\{\varepsilon_t = (\varepsilon_{1t}, \dots, \varepsilon_{nt})'\}$ is an i.i.d. sequence of n -dimensional random vectors with mean zero and covariance matrix $\Sigma_\varepsilon > 0$ such that $E|\varepsilon_{it}|^{2+\delta} < \infty$ for some $\delta > 0$.

We shall initialize (2) at $t = -k + 1, \dots, 0$ and allow the initial values $\{\eta_{-k+1}, \dots, \eta_0\}$ to be any random vectors including constants.

Substituting $\eta_t = y_t - \beta_0 - \beta_1 t - \cdots - \beta_q t^q$ into (2), we have

$$y_t = \gamma_0 + \gamma_1 t + \cdots + \gamma_q t^q + J_1 y_{t-1} + \cdots + J_k y_{t-k} + \varepsilon_t, \quad (3)$$

where γ_i ($i = 0, \dots, q$) are the functions of β_i and J_h ($i = 0, \dots, q$, $h = 1, \dots, k$). Note that if $d > 0$, the order of the polynomial trend in (3) might be lower than the order q of the polynomial in (1), i.e., $\gamma_{s+1} = \cdots = \gamma_q = 0$ for some $s < q$, depending on the structure of β_i 's and J_h 's. For example, let $q = 1$ and $d = 1$ in (1) and (2). Then, (3) becomes

$$y_t = \gamma_0 + \gamma_1 t + J_1 y_{t-1} + \cdots + J_k y_{t-k} + \varepsilon_t, \quad (4)$$

where $\gamma_0 = J(1)\beta_0 - J'(1)\beta_1$ and $\gamma_1 = J(1)\beta_1$ with $J(z) = I_n - J_1 z - \cdots - J_k z^k$. Hence, if $J(1)\beta_1 = 0$, we have $\gamma_1 = 0$. This is always true if the process is not cointegrated since then $J(1) = 0$, and this can also occur when the process is cointegrated because then $J(1)$ is of reduced rank $r < n$.

Suppose our interest is *not* in whether the process $\{y_t\}$ is integrated, cointegrated, or stationary, but in testing the hypothesis that is formulated as restrictions

$$\mathcal{H}_0: f(\phi) = 0 \quad (5)$$

on the parameter $\phi = \text{vec}(\Phi)$ of the model (3), where $\Phi = (J_1, \dots, J_k)$ and $f(\cdot)$ is an m -vector valued function satisfying the standard assumption:

(A2) $f(\cdot)$ is a twice continuously differentiable function with

$$\text{rank}(F(\cdot)) = m,$$

in a neighborhood of the true parameter value ϕ , where $F(\theta) = \partial f(\theta) / \partial \theta'$.

¹ In Section 4 we discuss a procedure to determine the lag length k when it is unknown.

To test the hypothesis (5) we consider estimating a levels VAR,

$$y_t = \hat{\gamma}_0 + \hat{\gamma}_1 t + \cdots + \hat{\gamma}_q t^q + \hat{J}_1 y_{t-1} + \cdots + \hat{J}_k y_{t-k} + \cdots + \hat{J}_p y_{t-p} + \hat{\varepsilon}_t, \quad (6)$$

by ordinary least squares (OLS), where $t = 1, \dots, T$, and $p \geq k + d$, i.e., we include at least d more lags than the true lag length k . Note that since the true values of $J_{k+1}, \dots, J_p$ are assumed to be zero, the parameter restrictions (5) do not involve them. Here and throughout the paper, a circumflex ($\hat{\cdot}$) denotes estimation by OLS.

Alternatively, if it is known that $\gamma_{s+1} = \cdots = \gamma_q = 0$ for some $s < q$ in (3), we may estimate²

$$y_t = \hat{\gamma}_0 + \hat{\gamma}_1 t + \cdots + \hat{\gamma}_s t^s + \hat{J}_1 y_{t-1} + \cdots + \hat{J}_k y_{t-k} + \cdots + \hat{J}_p y_{t-p} + \hat{\varepsilon}_t.$$

The asymptotics for the latter estimated equation is, in general, somewhat different³ from that for (6), but the results obtained below are unchanged. We shall deal with the estimated equation (6) in this paper.

Now, it is convenient to write (6) as

$$y_t = \hat{F} \tau_t + \hat{\Phi} x_t + \hat{\Psi} z_t + \hat{\varepsilon}_t, \quad (7)$$

where $\hat{F} = (\hat{\gamma}_0, \dots, \hat{\gamma}_q)$, $\tau_t = (1, t, \dots, t^q)'$, $x_t = (y'_{t-1}, \dots, y'_{t-k})'$, $z_t = (y'_{t-k-1}, \dots, y'_{t-p})'$, $\hat{\Phi} = (\hat{J}_1, \dots, \hat{J}_k)$, and $\hat{\Psi} = (\hat{J}_{k+1}, \dots, \hat{J}_p)$, or in the usual matrix notation:

$$Y' = \hat{F} \mathcal{T}' + \hat{\Phi} X' + \hat{\Psi} Z' + \hat{\varepsilon}',$$

where $\mathcal{T} = (\tau_1, \dots, \tau_T)'$, $X = (x_1, \dots, x_T)'$, and so on. With the estimated parameter $\hat{\phi} = \text{vec}(\hat{\Phi})$, we construct a standard Wald statistic $\mathcal{W}$ to test the hypothesis (5):

$$\mathcal{W} = f(\hat{\phi}) [F(\hat{\phi}) \{ \hat{\Sigma}_\varepsilon \otimes (X' Q X)^{-1} \} F(\hat{\phi})']^{-1} f(\hat{\phi}), \quad (8)$$

where $\hat{\Sigma}_\varepsilon = T^{-1} \hat{\varepsilon}' \hat{\varepsilon}$, $Q = Q_t - Q_t Z (Z' Q_t Z)^{-1} Z' Q_t$, and $Q_t = I_T - \mathcal{T} (\mathcal{T}' \mathcal{T})^{-1} \mathcal{T}'$ with I_T being the $T \times T$ identity matrix.

One of the objectives of this paper is to show:

Under the null hypothesis (5), the Wald statistic (8) has an asymptotic chi-square distribution with m degrees of freedom if $p \geq k + d$.

² If one postulates a data generating process such as (3) rather than starting from (1) and (2), one knows by assumption the order of the polynomial trend in the VAR representation for y_t , but not the order of the polynomial in y_t itself when $d > 0$. Alternatively, if one postulates the data-generating process (1) and (2) as we do in this paper, the order of the polynomial in y_t is known by assumption, but not the order of the polynomial in the VAR representation when $d > 0$.

³ See, for instance, Toda and Phillips (1993b) for the treatment of the case where the VAR process is $I(1)$ around a linear trend but time is not included in the estimation.

This implies that we can test general restrictions on the parameter matrices ($J_1, \dots, J_k$) of the data-generating process (DGP) using the usual chi-square critical values. All we need is to determine the maximal order of integration $d_{\max}$ which we suspect might occur in the model, and then to over-fit intentionally a levels VAR with additional $d_{\max}$ lags (i.e., $p = k + d_{\max}$). That is, we have to pay little attention to integration and cointegration properties of the DGP. For example, suppose we believe that the order of integration of y_t is at most two around a linear trend. Then, we should estimate the equation

$$y_t = \hat{\gamma}_0 + \hat{\gamma}_1 t + \hat{J}_1 y_{t-1} + \dots + \hat{J}_k y_{t-k} + \hat{J}_{k+1} y_{t-k-1} + \hat{J}_{k+2} y_{t-k-2} + \hat{\varepsilon}_t. \quad (9)$$

Under the null hypothesis (5), the Wald statistic (8) is asymptotically distributed as chi-square with the usual degrees of freedom, and this does not depend on whether y_t is stationary (around a linear trend), $I(1)$, or $I(2)$, or on whether y_t is cointegrated or not.

To prepare for the formal asymptotic analysis in the next section and to get some idea of why the Wald test (8) is valid asymptotically as chi-square criterion even if y_t is not stationary, we consider the following transformation of the model. For any positive integer j , define

$$H_j = \begin{bmatrix} I_n & I_n & I_n & \dots & I_n & I_n \\ 0 & I_n & I_n & \dots & I_n & I_n \\ 0 & 0 & I_n & \dots & I_n & I_n \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & I_n & I_n \\ 0 & 0 & 0 & \dots & 0 & I_n \end{bmatrix},$$

which is an $nj \times nj$ nonsingular matrix. We can easily check that the inverse matrix of H_j is given by

$$H_j^{-1} = \begin{bmatrix} I_n & -I_n & 0 & \dots & 0 & 0 \\ 0 & I_n & -I_n & \dots & 0 & 0 \\ 0 & 0 & I_n & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & I_n & -I_n \\ 0 & 0 & 0 & \dots & 0 & I_n \end{bmatrix}.$$

Then define, for any positive integer $u \leq p$, an $np \times np$ matrix

$$R_u = \begin{pmatrix} H_{p-u+1} & 0 \\ 0 & I_{n(u-1)} \end{pmatrix},$$

where $I_{n(u-1)}$ is the $n(u-1) \times n(u-1)$ identity matrix, and R_1 is taken as H_p . Furthermore, let

$$P_h = R_1 R_2 \cdots R_h,$$

for any positive integer $h \leq p$.

Now, define for $d \leq p - k$

$$(\Phi_d, \Psi_d) = (\Phi, \Psi) P_d \quad \text{and} \quad \begin{pmatrix} x_t^{(d)} \\ z_t^{(d)} \end{pmatrix} = P_d^{-1} \begin{pmatrix} x_t \\ z_t \end{pmatrix},$$

where $\Phi = (J_1, \dots, J_k)$, $\Psi = (J_{k+1}, \dots, J_p)$, $\Phi_d: n \times nk$, $\Psi_d: n \times n(p-k)$, $x_t^{(d)}: nk \times 1$, and $z_t^{(d)}: n(p-k) \times 1$, and we transform the DGP (3) as

$$\begin{aligned} y_t &= \Gamma \tau_t + (\Phi, \Psi) P_d P_d^{-1} \begin{pmatrix} x_t \\ z_t \end{pmatrix} + \varepsilon_t \\ &= \Gamma \tau_t + \Phi_d x_t^{(d)} + \Psi_d z_t^{(d)} + \varepsilon_t, \end{aligned} \quad (10)$$

where $\Gamma = (\gamma_0, \dots, \gamma_q)$. It is straightforward to see that

$$x_t^{(d)} = (\Delta^d y'_{t-1}, \dots, \Delta^d y'_{t-k}),$$

where $\Delta^d = (1 - L)^d$ with L being the lag operator such that $Ly_t = y_{t-1}$, and⁴

$$z_t^{(d)} = (\Delta^d y'_{t-k-1}, \dots, \Delta^d y'_{t-p+d}, \Delta^{d-1} y'_{t-p+d-1}, \dots, \Delta y'_{t-p+1}, y'_{t-p})'.$$

Let us define an $np \times nk$ matrix S ,

$$S = (I_{nk}, 0)'.$$

For any positive integer $u \leq p - k$ we have $R_u S = S H_k$, and hence

$$P_d S = R_1 R_2 \cdots R_d S = S H_k^d,$$

for $d \leq p - k$. Therefore, if $d \leq p - k$, we have⁵

$$\Phi_d = (\Phi, \Psi) P_d S = (\Phi, \Psi) S H_k^d = \Phi H_k^d.$$

Next, given $d \leq p - k$, define a function $g_d(\theta)$ by

$$g_d(\theta) = f((I_n \otimes H_k^{-d'}) \theta),$$

⁴ If $p = k + d$, $z_t^{(d)} = (\Delta^{d-1} y'_{t-k-1}, \dots, \Delta y'_{t-p+1}, y'_{t-p})'$.

⁵ The explicit forms of Φ_d and Ψ_d are given as follows: Write $\Phi_d = (J_1^{(d)}, \dots, J_k^{(d)})$ and $\Psi_d = (J_{k+1}^{(d)}, \dots, J_p^{(d)})$, and we have

$$J_i^{(d)} = \sum_{h=1}^d J_h^{(d-1)}, \quad i = 1, \dots, p - d + 1,$$

and $J_i^{(d)} = J_i^{(d-1)}$, $i = p - d + 2, \dots, p$, where $d \geq 1$ and $J_i^{(0)} = J_i$, $i = 1, \dots, p$.

where θ is an n^2k -vector. By construction, the restrictions

$$\mathcal{H}_0^{(d)}: g_d(\phi_d) = 0 \quad (11)$$

on the parameter ϕ_d , where $\phi_d = \text{vec}(\Phi_d)$, is equivalent to the restrictions (5). But, Φ_d is the coefficient matrix of the variables $x_t^{(d)} = (\Delta^d y_{t-1}^{(d)}, \dots, \Delta^d y_{t-k}^{(d)})'$, and from (1)

$$\Delta^d y_t = \beta_0^{(d)} + \beta_1^{(d)} t + \dots + \beta_{q-d}^{(d)} t^{q-d} + \Delta^d \eta_t,$$

for some constant vectors $\beta_i^{(d)}$ ($i = 0, \dots, q-d$).⁶ The vector $\Delta^d \eta_t$ is stationary if η_t is $I(d)$ and the deterministic polynomial trend is eliminated by the inclusion of τ_t in the estimation. Therefore, we would expect that the usual asymptotic theory should apply to the OLS estimator of Φ_d and hence to the Wald statistic for testing (11).⁷

In fact, the Wald statistic for testing (11) gives the same numerical value as the Wald statistic (8), as we now show. Let

$$G_d(\theta) = \partial g_d(\theta) / \partial \theta' = F((I_n \otimes H_k^{-d'}) \theta) (I_n \otimes H_k^{-d'}).$$

Lemma 1. Given d ($\leq p-k$), we may rewrite the Wald statistic (8) as

$$\mathcal{W} = g_d(\hat{\phi}_d)' [G_d(\hat{\phi}_d) \{ \hat{\Sigma}_\varepsilon \otimes (X_d' Q_d X_d)^{-1} \} G_d(\hat{\phi}_d)']^{-1} g_d(\hat{\phi}_d), \quad (12)$$

where $Q_d = Q_\tau - Q_\tau Z_d (Z_d' Q_\tau Z_d)^{-1} Z_d' Q_\tau$, $X_d = (x_1^{(d)}, \dots, x_T^{(d)})'$, $Z_d = (z_1^{(d)}, \dots, z_T^{(d)})'$, and $\hat{\phi}_d = \text{vec}(\hat{\Phi}_d)$ with

$$\hat{\Phi}_d = Y' Q_d X_d (X_d' Q_d X_d)^{-1}. \quad (13)$$

We note from (13) that $\hat{\Phi}_d$ is the OLS estimator of Φ_d in the estimated equation

$$y_t = \hat{\Gamma} \tau_t + \hat{\Phi}_d x_t^{(d)} + \hat{\Psi}_d z_t^{(d)} + \hat{\varepsilon}_t. \quad (14)$$

Moreover, it can easily be seen that the residual sum of squares from the regression (14) is numerically the same as that from the regression (7). Therefore,

⁶ If $d > q$, $\Delta^d y_t = \Delta^q \eta_t$.

⁷ Sims, Stock, and Watson (1990) observed from their analysis of a general linear model that Wald statistics for testing linear restrictions have asymptotic chi-square distributions if one can transform the model in such a way that the equivalent restrictions in the transformed model involve only the coefficients of stationary (mean zero) variables. Although, roughly speaking, the results of the present paper are implicitly included in this broad conclusion, we believe that those are worth mentioning explicitly. Furthermore, our asymptotic analysis in the next section somewhat differs from that of Sims, Stock, and Watson (1990). To conduct their asymptotic analysis they made assumptions for the transformed model in which different stochastic order components have been separated, and it is not in general clear what conditions on the original model satisfy those assumptions. In contrast, we start from a set of conditions on the original VAR model.

the Wald statistic for testing (5) in the levels estimation (7) gives the same numerical value as the Wald statistic for testing the hypothesis (11) in the regression (14).

Thus, the forgoing argument suggests that the Wald statistic (8) has an asymptotic chi-square distribution with the usual degrees of freedom, even if y_t might be an integrated or cointegrated process (provided that $p \geq k + d$).

3. The case where the variables are at most $I(2)$

To present a formal asymptotic analysis of the hypothesis testing in possibly nonstationary VAR's, we assume in this section that $\{y_t\}$ is at most $I(2)$ around a linear trend and may be cointegrated. We prove that the Wald statistic (8) with $q = 1$ and $p = k + 2$ has an asymptotic chi-square distribution with the usual degrees of freedom, invariant to whether $\{y_t\}$ may be $I(0)$, $I(1)$, or $I(2)$. We restrict our attention to the case of $d_{\max} = 2$ because explicit conditions under which VAR models are $I(1)$ or $I(2)$ have been worked out in the literature (Johansen, 1991, 1992) and because we expect most economic time series encountered in empirical studies to be at most $I(2)$.⁸ Setting $q = 1$ is just for simplicity and we can deal with a higher-order polynomial trend in an entirely analogous way.

Thus, the DGP we deal with in this section is (1) with $q = 1$ and (2) where $\{\eta_t\}$ may be $I(0)$, $I(1)$, or $I(2)$ and may be cointegrated. In particular, we adopt the conditions given in Johansen (1992) on the parameter matrices J_i 's, which ensure the process to be $I(1)$ or $I(2)$ and, in general, cointegrated. We first consider the conditions for the process to be $I(1)$. Write (2) as

$$\eta_t = J_1 \eta_{t-1} + \cdots + J_k \eta_{t-k} + J_{k+1} \eta_{t-k-1} + J_{k+2} \eta_{t-k-2} + \varepsilon_t, \quad (15)$$

where $J_{k+1} = J_{k+2} = 0$. We exclude explosive processes:

$$(A3) \quad |J(z)| = 0 \quad \text{implies} \quad |z| > 1 \quad \text{or} \quad z = 1, \quad \text{where} \quad J(z) = I_n - J_1 z - \cdots - J_{k+2} z^{k+2}.$$

Eq. (15) can be written in an ECM format:

$$\Delta \eta_t = J_1^* \Delta \eta_{t-1} + \cdots + J_{k+1}^* \Delta \eta_{t-k-1} + \Pi_2 \eta_{t-k-2} + \varepsilon_t, \quad (16)$$

⁸ But the asymptotic analysis given below should be extended in a straightforward manner to the case of an arbitrary $d_{\max}$ with an arbitrary order of cointegration. The extension is obvious especially if one is willing to start from convenient assumptions for the transformed model (10) rather than the original model (3). But an explicit set of conditions for VAR models to be $CI(d, b)$ is not known if $d > 2$.

where $J_i^\dagger = \sum_{h=1}^i J_h - I_n$ ($i = 1, \dots, k+1$) and $\Pi_2 = -J(1)$. Here Π_2 is a matrix such that

(A4) $\Pi_2 = AB'$ for some A and B , where A and B are $n \times r$ matrices of rank r ($0 < r < n$). If $\Pi_2 = 0$, we say $r = 0$.

Furthermore, we need

(A5) $A_\perp' \Pi_1 B_\perp$ is nonsingular, where $\Pi_1 = -J^\dagger(1)$ with $J^\dagger(z) = I_n - J_1^\dagger z - \dots - J_{k+1}^\dagger z^{k+1}$, and $A_\perp$ and $B_\perp$ are $n \times (n-r)$ matrices of rank $n-r$ such that $A'A_\perp = B'B_\perp = 0$. (If $r = 0$, we take $A_\perp = B_\perp = I_n$.)

Under assumptions (A3)–(A5), the process is $I(1)$, and is cointegrated if $r > 0$ (see Theorem 2 of Johansen, 1992).

Next, we consider the conditions for the process to be $I(2)$. Eq. (16) can further be rewritten as

$$\Delta^2 \eta_t = J_1^* \Delta^2 \eta_{t-1} + \dots + J_k^* \Delta^2 \eta_{t-k} + \Pi_1 \Delta \eta_{t-k-1} + \Pi_2 \eta_{t-k-2} + \varepsilon_t, \quad (17)$$

where $J_i^* = \sum_{h=1}^i J_h^\dagger - I_n$ ($i = 1, \dots, k$). In the $I(2)$ case we need, instead of (A5),

(A6) $\bar{A}_\perp' \Pi_1 \bar{B}_\perp = FG'$ for some F and G , where $\bar{A}_\perp = A_\perp (A_\perp' A_\perp)^{-1}$, $\bar{B}_\perp = B_\perp (B_\perp' B_\perp)^{-1}$, and F and G are $(n-r) \times s$ matrices of rank s ($0 < s < n-r$). If $\Pi_1 = 0$, we say $s = 0$.

Under (A3), (A4), (A6), and (2.8) of Johansen (1992) which is needed to prevent the process from being $I(3)$, the process is $I(2)$ and is cointegrated unless $r = s = 0$ (see Theorem 3 of Johansen, 1992).⁹

In the following, by saying $d = 1$ we mean that we are assuming (A3)–(A5). Similarly, when we say $d = 2$, we are assuming (A3), (A4), (A6), and (2.8) of Johansen (1992).

Since the order of integration of the process is assumed to be at most two, we include two extra lags in the estimated VAR, i.e., the estimated equation is (9). Formally, we prove the next theorem:

Theorem 1. Let $\mathcal{W}$ be the Wald statistic (8) with $q = 1$ and $p = k + 2$ for testing the hypothesis (5) based on the levels VAR estimation (9). If the process $\{y_t\}$ is

⁹ Johansen's (1992) formulation (1.2) of the ECM is slightly different from ours. But, assumptions (A4)–(A6) are equivalent to (1.3), (1.4), and (2.7), respectively, of Johansen (1992). Note in particular the relation that $(k+2) \Pi_2 = \Pi_1 + \Psi$ where Π_1 and Π_2 were defined above and Ψ is the matrix defined immediately above (1.2) of Johansen (1992).

stationary, $I(1)$, or $I(2)$, possibly around a linear trend in each case, then under the null hypothesis

$$\mathcal{W} \xrightarrow{d} \chi_m^2,$$

where y_t may be cointegrated if it is $I(1)$ or $I(2)$.

We consider mainly the case in which $d = 2$. If $d = 0$, i.e., y_t is stationary around the deterministic trend, then it is obvious that the conventional theory applies to the asymptotic analysis of the hypothesis test (8). The derivation of the limit distribution of the Wald statistic in the $I(1)$ case is analogous to that of the $I(2)$ case and will be discussed briefly later in this section.

Now, by Lemma 1 of the last section, the Wald statistic (8) with $q = 1$ and $p = k + 2$ may be written as

$$\mathcal{W} = g_2(\hat{\phi}_2)' [G_2(\hat{\phi}_2) \{ \hat{\Sigma}_\varepsilon \otimes (X_2' Q_2 X_2)^{-1} \} G_2(\hat{\phi}_2)']^{-1} g_2(\hat{\phi}_2), \quad (18)$$

where $g_2(\cdot)$, $\hat{\phi}_2$, X_2 , and so on are as defined in the last section with $d = 2$, $q = 1$, and $p = k + 2$. This is the Wald statistic for testing the hypothesis (11) with $d = 2$ in the regression

$$y_t = \hat{\Gamma} \tau_t + \hat{\Phi}_2 x_t^{(2)} + \hat{\Psi}_2 \tilde{z}_t^{(2)} + \hat{\varepsilon}_t, \quad (19)$$

where Γ , τ_t , $x_t^{(2)}$, and so on are as defined in Section 2 with $d = 2$, $q = 1$, and $p = k + 2$.

To obtain the limiting distribution of (18) we need a few preliminary results with regard to the stochastic component η_t in (1). Using the transformation matrix P_2 defined in the last section, we may write (2) as

$$\eta_t = \Phi_2 \tilde{x}_t^{(2)} + \Psi_2 \tilde{z}_t^{(2)} + \varepsilon_t,$$

where $\tilde{x}_t^{(2)} = (\Delta^2 \eta'_{t-1}, \dots, \Delta^2 \eta'_{t-k})'$ and $\tilde{z}_t^{(2)} = (\Delta \eta'_{t-k-1}, \eta'_{t-k-2})'$. Note that, by assumption, $\tilde{x}_t^{(2)}$ is stationary, and $\Delta \eta_{t-k-1}$ and η_{t-k-2} in $\tilde{z}_t^{(2)}$ are $I(1)$ and $I(2)$, respectively.

Next, we take into account the possibility of cointegration. By Theorem 3 of Johansen (1992) we can find a $2n \times 2n$ nonsingular matrix $C = (C_0, C_1, C_2)$, where C_0 , C_1 , and C_2 are $2n \times r_0$, $2n \times r_1$, and $2n \times r_2$ matrices, respectively, such that the r_0 -vector $C_0' \tilde{z}_t^{(2)}$ is $I(0)$, the r_1 -vector $C_1' \tilde{z}_t^{(2)}$ is $I(1)$ with no cointegration, and the r_2 -vector $C_2' \tilde{z}_t^{(2)}$ is $I(2)$ with no cointegration. (See the proof of Lemma 2 in the Appendix for the explicit form of C .) Note that, in general, C_0 involves so called polynomial cointegration vectors, i.e., some linear combinations of $\Delta \eta_{t-k-1}$ and η_{t-k-2} may be stationary.

Let

$$w_{0t} = \begin{pmatrix} \tilde{x}_t^{(2)} \\ C'_0 \tilde{z}_t^{(2)} \end{pmatrix} = \begin{pmatrix} w_{01t} \\ w_{02t} \end{pmatrix},$$

$$w_{1t} = C'_1 \tilde{z}_t^{(2)},$$

$$w_{2t} = C'_2 \tilde{z}_t^{(2)}.$$

To simplify the derivation below we assume that $\{\eta_{-k+1}, \dots, \eta_0\}$ are given the initial (joint) distribution such that w_{0t} , Δw_{1t} , and $\Delta^2 w_{2t}$ are stationary for all $t \geq 1$.¹⁰ Thus, let

$$w_t = (\varepsilon'_t, w'_{0t}, \Delta w'_{1t}, \Delta^2 w'_{2t})',$$

and we define for any t

$$\Sigma = E w_t w'_t,$$

$$\Lambda = \sum_{j=1}^{\infty} E w_t w'_{t+j},$$

$$\Omega = \Sigma + \Lambda + \Lambda'.$$

We partition Ω , Σ , and Λ conformably with w_t . For example,

$$\Sigma = \begin{pmatrix} \Sigma_{\varepsilon} & \Sigma_{\varepsilon 0} & \Sigma_{\varepsilon 1} & \Sigma_{\varepsilon 2} \\ \Sigma_{0\varepsilon} & \Sigma_0 & \Sigma_{01} & \Sigma_{02} \\ \Sigma_{1\varepsilon} & \Sigma_{10} & \Sigma_1 & \Sigma_{12} \\ \Sigma_{2\varepsilon} & \Sigma_{20} & \Sigma_{21} & \Sigma_2 \end{pmatrix}.$$

We start our asymptotic analysis with the next lemma.

Lemma 2

$$T^{-1} \sum_{t=1}^T w_{0t} w'_{0t} \xrightarrow{P} \Sigma_0 > 0$$

and

$$\begin{pmatrix} T^{-1/2} \sum_{t=1}^{[Ts]} \varepsilon_t \\ T^{-1/2} \sum_{t=1}^T (w_{0t} \otimes \varepsilon_t) \end{pmatrix} \xrightarrow{d} \begin{pmatrix} B_{\varepsilon}(s) \\ \xi \end{pmatrix} \begin{matrix} n \\ n(nk + r_0) \end{matrix},$$

where $B_{\varepsilon}(s)$ is a vector Brownian motion on $[0, 1]$ with covariance matrix $\Omega_{\varepsilon} = \Sigma_{\varepsilon}$, ξ is a normal random vector with mean zero and covariance matrix $\Sigma_0 \otimes \Sigma_{\varepsilon}$, and $B_{\varepsilon}(s)$ and ξ are independent.

¹⁰ Even if $\{\eta_{-k+1}, \dots, \eta_0\}$ are given an arbitrary distribution, w_{0t} , Δw_{1t} , and $\Delta^2 w_{2t}$ eventually become stationary. Hence, the asymptotic result below is unchanged.

The next lemma summarizes the asymptotic behavior of the sample moment matrices we use in deriving the limit distribution of the Wald statistic (18).

Lemma 3

- (i) (a) $T^{-1/2} \sum_{t=1}^T \varepsilon_t \xrightarrow{d} B_\varepsilon(1),$
 (b) $T^{-1/2} \sum_{t=1}^T w_{0t} \xrightarrow{d} B_0(1),$
 (c) $T^{-3/2} \sum_{t=1}^T w_{1t} \xrightarrow{d} \int_0^1 B_1(s) ds,$
 (d) $T^{-5/2} \sum_{t=1}^T w_{2t} \xrightarrow{d} \int_0^1 \bar{B}_2(s) ds,$
- (ii) (a) $T^{-3/2} \sum_{t=1}^T t\varepsilon_t \xrightarrow{d} \int_0^1 s dB_\varepsilon(s),$
 (b) $T^{-3/2} \sum_{t=1}^T tw_{0t} \xrightarrow{d} \int_0^1 s dB_0(s),$
 (c) $T^{-5/2} \sum_{t=1}^T tw_{1t} \xrightarrow{d} \int_0^1 s B_1(s) ds,$
 (d) $T^{-7/2} \sum_{t=1}^T tw_{2t} \xrightarrow{d} \int_0^1 s \bar{B}_2(s) ds,$
- (iii) (a) $T^{-1} \sum_{t=1}^T w_{1t} \varepsilon'_t \xrightarrow{d} \int_0^1 B_1(s) dB_\varepsilon(s)',$
 (b) $T^{-1} \sum_{t=1}^T w_{1t} w'_{0t} \xrightarrow{d} \int_0^1 B_1(s) dB_0(s)' + \Sigma_{10} + A_{10},$
 (c) $T^{-2} \sum_{t=1}^T w_{1t} w'_{1t} \xrightarrow{d} \int_0^1 B_1(s) B_1(s)' ds,$
- (iv) (a) $T^{-2} \sum_{t=1}^T w_{2t} \varepsilon'_t \xrightarrow{d} \int_0^1 \bar{B}_2(s) dB_\varepsilon(s)',$
 (b) $T^{-2} \sum_{t=1}^T w_{2t} w'_{0t} \xrightarrow{d} \int_0^1 \bar{B}_2(s) dB_0(s)',$
 (c) $T^{-3} \sum_{t=1}^T w_{2t} w'_{1t} \xrightarrow{d} \int_0^1 \bar{B}_2(s) B_1(s)' ds,$
 (d) $T^{-4} \sum_{t=1}^T w_{2t} w'_{2t} \xrightarrow{d} \int_0^1 \bar{B}_2(s) \bar{B}_2(s)' ds,$

where

$$\begin{pmatrix} B_e(s) \\ B_0(s) \\ B_1(s) \\ B_2(s) \end{pmatrix} \begin{pmatrix} n \\ nk + r_0 \\ r_1 \\ r_2 \end{pmatrix}$$

is an $(n + nk + r_0 + r_1 + r_2)$ -vector Brownian motion whose covariance matrix is Ω with $\Omega_1 > 0$ and $\Omega_2 > 0$, and

$$\bar{B}_2(s) = \int_0^s B_2(u) du.$$

Now we are ready to analyze the asymptotics of $\mathcal{W}$ in (18). First, note that from (1) with $q = 1$ we have

$$y_t = \beta_0 + \beta_1 t + \eta_t,$$

$$\Delta y_t = \beta_1 + \Delta \eta_t, \quad \Delta^2 y_t = \Delta^2 \eta_t.$$

It follows that $Q_\tau X_2 = Q_\tau \tilde{X}_2$ and $Q_\tau Z_2 = Q_\tau \tilde{Z}_2$ where $\tilde{X}_2 = (\tilde{x}_1^{(2)}, \dots, \tilde{x}_T^{(2)})'$ and $\tilde{Z}_2 = (\tilde{z}_1^{(2)}, \dots, \tilde{z}_T^{(2)})'$. Hence, from $\hat{\Phi}_2 = Y' Q_2 X_2 (X_2' Q_2 X_2)^{-1}$ and $Y' = \Gamma \mathcal{T}' + \Phi_2 X_2' + \Psi_2 Z_2' + \mathcal{E}'$, we have

$$\hat{\Phi}_2 - \Phi_2 = \mathcal{E}' Q_2 X_2 (X_2' Q_2 X_2)^{-1} = \mathcal{E}' Q_2 \tilde{X}_2 (\tilde{X}_2' Q_2 \tilde{X}_2)^{-1},$$

where

$$Q_2 = Q_\tau - Q_\tau \tilde{Z}_2 (\tilde{Z}_2' Q_\tau \tilde{Z}_2)^{-1} \tilde{Z}_2' Q_\tau.$$

Moreover, defining $V = \tilde{Z}_2 C$, Q_2 may further be written as

$$Q_2 = Q_\tau - Q_\tau V (V' Q_\tau V)^{-1} V' Q_\tau.$$

Note that if we use the notation of Lemma 3, $\tilde{X}_2 = W_{01}$ and $V = (W_{02}, W_1, W_2)$ where $W_1 = (w_{11}, \dots, w_{1T})'$ and so forth. Thus, Lemma 4 below is needed to obtain the limit distribution of the OLS estimator of Φ_2 . Let

$$\Sigma_0 = \begin{pmatrix} \Sigma_0^{11} & \Sigma_0^{12} \\ \Sigma_0^{21} & \Sigma_0^{22} \end{pmatrix},$$

where Σ_0 is partitioned conformably with $w_{0t} = (w'_{01t}, w'_{02t})' = (\tilde{x}_t^{(2)'} , (C'_0 \tilde{z}_t^{(2)})')'$. Also, let

$$\xi = \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix},$$

where ξ is partitioned conformably with $w_{0t} \otimes \varepsilon_t = (w'_{01t} \otimes \varepsilon'_t, w'_{02t} \otimes \varepsilon'_t)'$. Furthermore, let

$$D_T = \begin{pmatrix} T^{1/2} I_{r_0} & 0 & 0 \\ 0 & T I_{r_1} & 0 \\ 0 & 0 & T^2 I_{r_2} \end{pmatrix}.$$

Lemma 4

$$(i) \quad (a) \quad T^{-1} \tilde{X}'_2 Q_\tau \tilde{X}_2 \xrightarrow{P} \Sigma_0^{11},$$

$$(b) \quad D_T^{-1} V' Q_\tau V D_T^{-1} \xrightarrow{d} \begin{pmatrix} \Sigma_0^{22} & 0 \\ 0 & \int_0^1 \underline{B}_*(s) \underline{B}_*(s)' ds \end{pmatrix},$$

$$(c) \quad T^{-1/2} \tilde{X}'_2 Q_\tau V D_T^{-1} \xrightarrow{P} (\Sigma_0^{12}, 0),$$

$$(ii) \quad (a) \quad T^{-1/2} \text{vec}(\tilde{X}'_2 Q_\tau \mathcal{E}) \xrightarrow{d} \xi_1,$$

$$(b) \quad (D_T^{-1} \otimes I_n) \text{vec}(V' Q_\tau \mathcal{E}) \xrightarrow{d} \begin{pmatrix} \xi_2 \\ \text{vec} \left(\int_0^1 \underline{B}_*(s) d\mathcal{B}_\varepsilon(s)' \right) \end{pmatrix},$$

where $B_*(s) = (B_1(s)', \bar{B}_2(s)')'$ and

$$\underline{B}_*(s) = B_*(s) - \int_0^1 B_*(u) \delta(u)' du \left(\int_0^1 \delta(u) \delta(u)' du \right)^{-1} \delta(s),$$

with $\delta(s) = (1, s)'$.

The results in Lemma 4 are immediate consequences of Lemma 3.

Now it follows from Lemma 4 that

$$T^{-1} \tilde{X}'_2 Q_2 \tilde{X}_2 \xrightarrow{P} \Sigma_0^{1 \cdot 2}, \quad (20)$$

where $\Sigma_0^{1 \cdot 2} = \Sigma_0^{11} - \Sigma_0^{12}(\Sigma_0^{22})^{-1}\Sigma_0^{21}$, and that¹¹

$$\begin{aligned}\sqrt{T}(\hat{\phi}_2 - \phi_2) &= \sqrt{T} \text{vec}(\hat{\Phi}_2 - \Phi_2) \\ &= \sqrt{T} K_{n, nk} \text{vec}((\tilde{X}'_2 Q_2 \tilde{X}_2)^{-1} \tilde{X}'_2 Q_2 \mathcal{E}) \\ &= K_{n, nk} \{ (T^{-1} \tilde{X}'_2 Q_2 \tilde{X}_2)^{-1} \otimes I_n \} T^{-1/2} \text{vec}(\tilde{X}'_2 Q_2 \mathcal{E}) \\ &\stackrel{d}{\rightarrow} K_{n, nk} \{ (\Sigma_0^{1 \cdot 2})^{-1} \otimes I_n \} K_{nk, n} K_{n, nk} (\xi_1 - \{ \Sigma_0^{12} (\Sigma_0^{22})^{-1} \otimes I_n \} \xi_2) \\ &\equiv N(0, \Sigma_\varepsilon \otimes (\Sigma_0^{1 \cdot 2})^{-1}),\end{aligned}$$

where vec is the row-stacking operator and $K_{l, m}$ is the commutation matrix such that $K_{l, m} \text{vec}(M') = \text{vec}(M)$ for an $l \times m$ matrix M , $K'_{l, m} = K_{l, m}^{-1} = K_{m, l}$ and $K_{g, l}(M_1 \otimes M_2)K_{m, h} = M_2 \otimes M_1$ for an $l \times m$ matrix M_1 and an $g \times h$ matrix M_2 . Therefore, since $g_2(\cdot)$ clearly satisfies the same qualification as (A2) for $f(\cdot)$, by the standard argument of applying a Taylor series expansion to $g_2(\cdot)$,

$$\sqrt{T}g_2(\hat{\phi}_2) \stackrel{d}{\rightarrow} N(0, G_2(\phi_2) \{ \Sigma_\varepsilon \otimes (\Sigma_0^{1 \cdot 2})^{-1} \} G_2(\phi_2)') \quad (21)$$

under the null hypothesis $g_2(\phi_2) = 0$. Next, by the consistency of $\hat{\Phi}_2$

$$G_2(\hat{\phi}_2) \xrightarrow{p} G_2(\phi_2). \quad (22)$$

Furthermore, it easily follows from the consistency¹² of $\hat{\Phi}_2$ and $\hat{\Psi}_2$ (or $\hat{\Phi}$ and $\hat{\Psi}$) that

$$\hat{\Sigma}_\varepsilon \xrightarrow{p} \Sigma_\varepsilon. \quad (23)$$

Thus, combining (20)–(23) we deduce that

$$\mathcal{W} \xrightarrow{d} \chi_m^2.$$

¹¹ If the process is not cointegrated, there is no stationary component in $\tilde{z}_t^{(2)}$, so we have $T^{-1} \tilde{X}'_2 Q_2 \tilde{X}_2 \xrightarrow{d} \Sigma_0$ and $\sqrt{T}(\hat{\phi}_2 - \phi_2) \xrightarrow{d} N(0, \Sigma_\varepsilon \otimes \Sigma_0^{-1})$.

¹² The consistency of OLS estimators in linear regressions with integrated processes is a well-known fact. Hence, the proof of the consistency of $\hat{\Psi}_2$ (or $\hat{\Psi}$) is omitted. It can easily be proved using Lemma 4.

To prove that $\mathcal{W}$ converges in distribution to a chi-square random variable with m degrees of freedom in the case of $d = 1$, we use the fact that the Wald statistic (8) is numerically the same as

$$\mathcal{W} = g_1(\hat{\phi}_1)' [G_1(\hat{\phi}_1) \{ \hat{\mathcal{L}}_e \otimes (X_1' Q_1 X_1)^{-1} \} G_1(\hat{\phi}_1)']^{-1} g_1(\hat{\phi}_1)$$

in the estimated system

$$y_t = \hat{F}\tau_t + \hat{\phi}_1 x_t^{(1)} + \hat{\Psi}_1 z_t^{(1)} + \hat{\varepsilon}_t,$$

where $x_t^{(1)} = (\Delta y_{t-1}, \dots, \Delta y_{t-k})'$ and $z_t^{(1)} = (\Delta y_{t-k-1}, y_{t-k-2})'$. Using the matrices B and $B_\perp$ introduced in (A4) and (A5), define

$$w_t = (\varepsilon_t', w_{0t}', \Delta w_{1t}')',$$

where $w_{1t} = B_\perp' \eta_{t-k-2}$, and

$$w_{0t} = \begin{pmatrix} \hat{x}_t^{(1)} \\ \Delta \eta_{t-k-1} \\ B' \eta_{t-k-2} \end{pmatrix},$$

with $\hat{x}_t^{(1)} = (\Delta \eta_{t-1}', \dots, \Delta \eta_{t-k}')'$. Then, given Lemma 1 and Lemma 2 of Toda and Phillips (1993a), the rest of the proof for the $I(1)$ case is entirely analogous to the $I(2)$ case.

As mentioned before, it is obvious that $\mathcal{W} \xrightarrow{d} \chi_m^2$ in the case of $d = 0$. This completes the proof of Theorem 1.

Remark 1. Note that if $d = 2$ and if the cointegrating matrix C_0 were known, we could estimate the 'ECM' (still including two extra lags)

$$\Delta^2 y_t = \hat{F}\tau_t + \hat{J}_1^* \Delta^2 y_{t-1} + \dots + \hat{J}_k^* \Delta^2 y_{t-k} + \hat{D}_0 C_0' z_t^{(2)} + \hat{\varepsilon}_t$$

instead of (9) or equivalently (19). (See the proof of Lemma 2 in the Appendix for the explicit form of D_0 .) Let $\Phi_\star = (J_1^\star, \dots, J_k^\star)$ and $\hat{\Phi}_\star$ be the OLS estimator of $\Phi_\star$ in the last equation. Then, it is easy to see that the restrictions equivalent to (11) can be imposed on $\hat{\Phi}_\star$ and that the limiting distribution of $\sqrt{T}(\hat{\Phi}_\star - \Phi_\star)$ is exactly the same as that of $\sqrt{T}(\hat{\Phi}_2 - \Phi_2)$. Therefore, apart from the inefficiency that arises from intentionally over-fitting the VAR, there is no additional loss of asymptotic efficiency in taking no account of the cointegrating relations explicitly in the estimation. The same conclusion also applies in the case of $d = 1$.

Remark 2. Wald test (8) is clearly consistent. Suppose, for example, $\{y_t\}$ is $I(2)$ and consider the alternative hypothesis

$$\mathcal{H}_1: f(\phi) = \delta \neq 0,$$

or equivalently

$$\mathcal{H}_1^{(2)}: g(\phi_2) = \delta \neq 0.$$

Then, an analogous argument to that leading to (21) gives

$$\sqrt{T}(g_2(\hat{\phi}_2) - \delta) \xrightarrow{d} N(0, G_2(\phi_2) \{ \Sigma_\varepsilon \otimes (\Sigma_0^{1/2})^{-1} \} G_2(\phi_2)'),$$

and (22) and (23) still hold. Hence, under the alternative hypothesis $\mathcal{H}_1^{(2)}$, we have for any positive number c

$$\begin{aligned} \Pr\{\mathcal{W} > c\} &= \Pr\{[\sqrt{T}(g_2(\hat{\phi}_2) - \delta) + \sqrt{T}\delta]' \\ &\quad \times [G_2(\hat{\phi}_2) \{ \hat{\Sigma}_\varepsilon \otimes (T^{-1}X_2'Q_2X_2)^{-1} \} G_2(\hat{\phi}_2)']^{-1} \\ &\quad \times [\sqrt{T}(g_2(\hat{\phi}_2) - \delta) + \sqrt{T}\delta] > c\} \\ &\rightarrow 1 \quad \text{as } T \rightarrow \infty. \end{aligned}$$

The same conclusion obviously holds when $d = 0$ or $d = 1$.

4. The selection of lag length

In the last two sections we assumed that the true lag length k of the model is known *a priori*. But it rarely is the case in practice. In this section we shall show that a lag selection procedure that is commonly employed for stationary VAR's is valid even for VAR's with integrated or cointegrated processes¹³ as far as $k \geq d$.

Since the formal asymptotic analysis of this problem is entirely similar to that of the hypothesis testing discussed in the last section, we present only an intuitive argument in the framework of the general model formulated in Section 2. Thus, let the n -vector time series $\{y_t\}_{t=-k+1}^\infty$ be generated by (1) and (2) where $\{\eta_t\}$ is $I(d)$ or $CI(d, b)$. We write the DGP as

$$y_t = \gamma_0 + \gamma_1 t + \cdots + \gamma_q t^q + J_1 y_{t-1} + \cdots + J_k y_{t-k} + \cdots + J_p y_{t-p} + \varepsilon_t, \quad (24)$$

where $J_{k+1} = \cdots = J_p = 0$ ($p \geq k+1$). Suppose we wish to test the hypothesis

$$\mathcal{H}_0^*: J_{m+1} = \cdots = J_p = 0, \quad (25)$$

where $k \leq m \leq p-1$, in the estimated system

$$y_t = \hat{\gamma}_0 + \hat{\gamma}_1 t + \cdots + \hat{\gamma}_q t^q + \hat{J}_1 y_{t-1} + \cdots + \hat{J}_p y_{t-p} + \hat{\varepsilon}_t. \quad (26)$$

¹³ Sims, Stock, and Watson (1990) showed in their Example 1 of Section 6 that the procedure we discuss below is valid in trivariate VAR's with $I(1)$ processes.

Write (26) as

$$y_t = \hat{\Gamma} \tau_t + \hat{\Psi} z_t + \hat{\Phi} x_t + \hat{\varepsilon}_t,$$

where $\hat{\Gamma} = (\hat{\gamma}_0, \dots, \hat{\gamma}_q)$, $\tau_t = (1, t, \dots, t^q)'$, $z_t = (y'_{t-1}, \dots, y'_{t-m})'$, $x_t = (y'_{t-m-1}, \dots, y'_{t-p})'$, $\hat{\Psi} = (\hat{J}_1, \dots, \hat{J}_m)$, and $\hat{\Phi} = (\hat{J}_{m+1}, \dots, \hat{J}_p)$ or in the corresponding matrix notation

$$Y' = \hat{\Gamma} \mathcal{T}' + \hat{\Psi} Z' + \hat{\Phi} X' + \hat{\varepsilon}'.$$

With the estimated parameter $\hat{\phi} = \text{vec}(\hat{\Phi})$, we construct the Wald statistic $\mathcal{W}^\dagger$ to test the hypothesis (25):

$$\mathcal{W}^\dagger = \hat{\phi}' [\hat{\Sigma}_\varepsilon \otimes (X' Q X)^{-1}]^{-1} \hat{\phi}, \quad (27)$$

where $\hat{\Sigma}_\varepsilon = T^{-1} \hat{\varepsilon}' \hat{\varepsilon}$, $Q = Q_\tau - Q_\tau Z' (Z' Q_\tau Z)^{-1} Z' Q_\tau$, and $Q_\tau = I_T - \mathcal{T} (\mathcal{T}' \mathcal{T})^{-1} \mathcal{T}'$ as before.

Now, what we want to show is the following:

Under the null hypothesis (25), the Wald statistic (27) has an asymptotic chi-square distribution with $n^2(p-m)$ degrees of freedom if $m \geq d$.

To see this, as in Section 2, we transform the model using some matrices. For any positive integer $u \leq p$, define the new R_u matrix

$$R_u = \begin{pmatrix} I_{n(u-1)} & 0 \\ 0 & -H'_{p-u+1} \end{pmatrix},$$

which is an $np \times np$ matrix, where H_j ($j = 1, \dots, p$) were defined in Section 2, and R_1 is taken as $-H'_p$. Further, let $P_h = R_1 R_2 \dots R_h$ for any positive integer $h \leq p$ as before. Then, define

$$(\Psi_d, \Phi_d) = (\Psi, \Phi) P_d \quad \text{and} \quad \begin{pmatrix} z_t^{(d)} \\ x_t^{(d)} \end{pmatrix} = P_d^{-1} \begin{pmatrix} z_t \\ x_t \end{pmatrix},$$

where $\Psi = (J_1, \dots, J_m)$, $\Phi = (J_{m+1}, \dots, J_p)$, $\Psi_d: n \times nm$, $\Phi_d: n \times n(p-m)$, $z_t^{(d)}: nm \times 1$, and $x_t^{(d)}: n(p-m) \times 1$, and we transform (24) as

$$\begin{aligned} y_t &= \Gamma \tau_t + (\Psi, \Phi) P_d P_d^{-1} \begin{pmatrix} z_t \\ x_t \end{pmatrix} + \varepsilon_t \\ &= \Gamma \tau_t + \Psi_d z_t^{(d)} + \Phi_d x_t^{(d)} + \varepsilon_t. \end{aligned} \quad (28)$$

It is easy to check that for $d \leq m$

$$x_t^{(d)} = (\Delta^d y'_{t-m+d-1}, \dots, \Delta^d y'_{t-p+d})'$$

and¹⁴

$$z_t^{(d)} = (-y'_{t-1}, -\Delta y'_{t-1}, \dots, -\Delta^{d-1} y'_{t-1}, \Delta^d y'_{t-1}, \Delta^d y'_{t-2}, \dots, \Delta^d y'_{t-m+d})'.$$

Note also that for $u \leq m$,

$$R_u S = -SH'_{p-m},$$

where $S = (0, I_{n(p-m)})'$ is an $np \times n(p-m)$ matrix. Hence¹⁵

$$\Phi_d = (\Psi, \Phi) P_d S = (\Psi, \Phi) SH'^d_{p-m} (-1)^d = \Phi H'^d_{p-m} (-1)^d,$$

for $d \leq m$. Therefore, the hypothesis (25) is equivalent to

$$\mathcal{H}_0^{\dagger(d)}: \Phi_d = 0 \quad (29)$$

in the model (28). But, Φ_d is the coefficient matrix of the vector $x_t^{(d)} = (\Delta^d y'_{t-m+d-1}, \dots, \Delta^d y'_{t-p+d})'$. This vector is stationary around a $(q-d)$ th-order polynomial trend, which is eliminated by the inclusion of τ_t in the estimation. Consequently, the usual asymptotics apply to the OLS estimator of Φ_d and hence to the Wald statistic for testing (29).

As in Section 2, we next see that the Wald statistic for testing (29) in the regression (31) below is, in fact, numerically the same as that for testing (25) in (26). By the argument similar to that in the proof of Lemma 1, given $d \leq m$, we can rewrite the Wald statistic (27) as

$$\mathcal{W}^{\dagger} = \hat{\phi}'_d [\hat{\Sigma}_\varepsilon \otimes (X'_d Q_d X_d)^{-1}]^{-1} \hat{\phi}_d, \quad (30)$$

where $\hat{\phi}_d = \text{vec}(\hat{\Phi}_d)$ with $\hat{\Phi}_d = \hat{\Phi} H'^d_{p-m} (-1)^d = Y' Q_d X_d (X'_d Q_d X_d)^{-1}$, $Q_d = Q_\tau - Q_\tau Z_d (Z'_d Q_\tau Z_d)^{-1} Z'_d Q_\tau$, $X_d = (x_1^{(d)}, \dots, x_T^{(d)})'$, and $Z_d = (z_1^{(d)}, \dots, z_T^{(d)})'$. As before, $\hat{\Phi}_d$ is the OLS estimator of Φ_d in the regression

$$y_t = \hat{f} \tau_t + \hat{\Psi}_d z_t^{(d)} + \hat{\Phi}_d x_t^{(d)} + \hat{\varepsilon}_t. \quad (31)$$

Again, it can easily be seen that the residual sum of squares from the regression (31) is numerically the same as that from the regression (26). Thus, we conclude that the Wald statistic for testing (25) in the levels estimation (26) gives the same numerical value as the Wald statistic for testing the hypothesis (29) in the estimated equation (31). Therefore, the usual asymptotic theory applies to the hypothesis testing (25) even if the VAR process is integrated or cointegrated (provided that $m \geq d$).

¹⁴ If $m = d$, $z_t^{(d)} = (-y'_{t-1}, -\Delta y'_{t-1}, \dots, -\Delta^{d-1} y'_{t-1})'$.

¹⁵ Writing $\Phi_d = (J_{m+1}^{(d)}, \dots, J_p^{(d)})$ and $\Psi_d = (J_1^{(d)}, \dots, J_m^{(d)})$, the explicit forms of Φ_d and Ψ_d are given by

$$J_i^{(d)} = \sum_{h=i}^p J_h^{(d-1)}, \quad i = d, \dots, p,$$

and $J_i^{(d)} = J_i^{(d-1)}$, $i = 1, \dots, d-1$, where $d \geq 1$ and $J_i^{(0)} = J_i$, $i = 1, \dots, p$.

Note that if $k \geq d$, then $m \geq k \geq d$. Hence, the usual lag selection procedure is valid even in VAR's with integrated processes if the orders d of the integration of the processes do not exceed the true lag lengths k .¹⁶ That is, by testing the significance of $J_{k+1}, \dots, J_p$ for some $p > k$, we can choose the correct lag length k (with a desired significance level), at least asymptotically. By the same argument as that in Remark 2 of the previous section, this test procedure is clearly consistent (i.e., it does not under-estimate the lag length asymptotically).

The following example illustrates why we need the condition $k \geq d$. Suppose $k = 2$ and $p = 3$. If $d = 1$, (28) becomes

$$y_t = \Gamma \tau_t - J_1^{(1)} y_{t-1} + J_2^{(1)} \Delta y_{t-1} + J_3^{(1)} \Delta y_{t-2} + \varepsilon_t,$$

where $J_i^{(1)} = -\sum_{h=i}^3 J_h$ ($i = 1, 2, 3$). If $d = 2$,

$$y_t = \Gamma \tau_t - J_1^{(2)} y_{t-1} - J_2^{(2)} \Delta y_{t-1} + J_3^{(2)} \Delta^2 y_{t-1} + \varepsilon_t,$$

where $J_i^{(2)} = -\sum_{h=i}^3 J_h^{(1)}$ ($i = 2, 3$) and $J_1^{(2)} = J_1^{(1)}$. Note that $J_3^{(1)} = -J_3$ and $J_3^{(2)} = J_3$, so the restriction $\mathcal{H}_0^*: J_3 = 0$ can be expressed as the restriction on the coefficient matrix of a (trend) stationary vector in the transformed model if $d = 1$ or $d = 2$. But if $k = m = 2$ and $d = 3$, this is not the case.

This condition that $k \geq d$ should not be restrictive in practice since the orders of integration of time series we encounter in most empirical studies would be one or two. If $d = 1$, the lag selection procedure is always valid, at least asymptotically, since $k \geq 1 = d$. If $d = 2$, the procedure is asymptotically valid unless $k = 1$.

So far in this section we have focused on Wald tests of the significance of the lagged vectors. But for that purpose LR tests are probably used more often in practice. Therefore, it is perhaps worth noting that LR tests can also be employed in the usual way. It should be clear that Wald and LR tests are asymptotically equivalent in the present situation.

5. Conclusion

This paper has shown how we can estimate levels VAR's and test general restrictions on the parameter matrices even if the processes may be integrated or cointegrated of an arbitrary order; we can apply the usual lag selection procedure discussed in Section 4 to a possibly integrated or cointegrated VAR (as far as the order of integration of the process does not exceed the true lag length of the model). Having chosen a lag length k , we then estimate a $(k + d_{\max})$ -th-order VAR where $d_{\max}$ is the maximal order of integration that we suspect might occur

¹⁶ If the process is not cointegrated, d cannot exceed k , but if cointegrated, d can be greater than k .

in the process. The coefficient matrices of the last $d_{\max}$ lagged vectors in the model are ignored (since these are regarded as zeros), and we can test linear or nonlinear restrictions on the first k coefficient matrices using the standard asymptotic theory.

We proposed a simple way to test economic hypotheses expressed as restrictions on the parameters of VAR models without pretests for a unit root(s) and a cointegrating rank(s). Hypothesis tests such as (5) in levels VAR's, in general, involve not only nonstandard distributions but also nuisance parameters if the processes are integrated or cointegrated, and critical values for the tests cannot conveniently be tabulated. So the usual way to proceed is formulating equivalent ECM's in which most hypothesis testing can be conducted using the standard asymptotic theory. But this requires pretests of a unit root and cointegrating rank, which one may wish to avoid if the cointegrating relation itself is not one's interest since those tests are known to have low power. Hence, our simple method of adding extra lags intentionally in the estimation should be very useful in practice.

Of course, our approach is inefficient and suffers some loss of power since we intentionally over-fit VAR's. The relative inefficiency depends on a particular model employed. If, for instance, a VAR system has many variables and the true lag length is one, then the inefficiency caused by adding even one extra lag might be relatively big. On the other hand, if a VAR system has a small number of variables and long lag length as is often the case in practice, then the inefficiency caused by adding a few more lags might be relatively small. If the latter is the case, the pretest biases associated with the unit root and cointegration tests could be much more serious.

We emphasize, however, that we are *not* suggesting that our method should totally replace the conventional hypothesis testing that are conditional on the estimation of unit roots and cointegrating ranks. It should rather be regarded as complementing the pretesting method that may suffer serious biases in some cases.

Similarly, though the argument in Sections 2 and 3 is also applicable to Dickey–Fuller-type unit root tests and presumably to Johansen-type tests for cointegration,¹⁷ it is not recommended to apply our method to these problems. Since the limiting distributions for the unit root and cointegration tests (with the correct specification of the lag length) are free of nuisance parameters and the critical values are already known, there is no incentive to introduce the inefficiency by adding an extra lag even though it brings the problem within the scope of the conventional asymptotic theory.

Finally, the deterministic trends we considered in this paper are simple polynomials in time. It would be straightforward to extend the analysis so that

¹⁷ That is, adding an extra lag makes it possible to express the unit root or cointegration hypothesis as restrictions on stationary variables.

we can allow for more general deterministic trends such as those considered in Park (1992). Moreover, seasonal dummies may also be incorporated into the model in such a way as Johansen (1991).

Appendix

Proof of Lemma 1

Using $P_d S = S H_k^d$ we have

$$\begin{aligned} H_k^d (X' Q X)^{-1} H_k^d &= H_k^d S' \left\{ \begin{pmatrix} X' \\ Z' \end{pmatrix} Q_\tau(X, Z) \right\}^{-1} S H_k^d \\ &= S' \left\{ P_d^{-1} \begin{pmatrix} X' \\ Z' \end{pmatrix} Q_\tau(X, Z) P_d^{-1'} \right\}^{-1} S \\ &= S' \left\{ \begin{pmatrix} X'_d \\ Z'_d \end{pmatrix} Q_\tau(X_d, Z_d) \right\}^{-1} S \\ &= (X'_d Q_d X_d)^{-1}. \end{aligned}$$

Therefore

$$\begin{aligned} \mathcal{W} &= f((I_n \otimes H_k^{-d'}) \hat{\phi}_d)' \\ &\quad \times [F((I_n \otimes H_k^{-d'}) \hat{\phi}_d) \{ \hat{\Sigma}_\varepsilon \otimes (X' Q X)^{-1} \} F((I_n \otimes H_k^{-d'}) \hat{\phi}_d)]^{-1} \\ &\quad \times f((I_n \otimes H_k^{-d'}) \hat{\phi}_d) \\ &= g_d(\hat{\phi}_d)' [G_d(\hat{\phi}_d) \{ \hat{\Sigma}_\varepsilon \otimes H_k^d (X' Q X)^{-1} H_k^d \} G_d(\hat{\phi}_d)]^{-1} g_d(\hat{\phi}_d) \\ &= g_d(\hat{\phi}_d)' [G_d(\hat{\phi}_d) \{ \hat{\Sigma}_\varepsilon \otimes (X'_d Q_d X_d)^{-1} \} G_d(\hat{\phi}_d)]^{-1} g_d(\hat{\phi}_d), \end{aligned}$$

where we have defined $\hat{\phi}_d = \text{vec}(\hat{\Phi}_d)$ with $\hat{\Phi}_d = \hat{\Phi} H_k^d$. Furthermore,

$$\begin{aligned} \hat{\Phi}_d &= \hat{\Phi} H_k^d \\ &= Y' Q X (X' Q X)^{-1} H_k^d \\ &= Y' Q_\tau(X, Z) \left\{ \begin{pmatrix} X' \\ Z' \end{pmatrix} Q_\tau(X, Z) \right\}^{-1} S H_k^d \\ &= Y' Q_\tau(X, Z) P_d^{-1'} \left\{ P_d^{-1} \begin{pmatrix} X' \\ Z' \end{pmatrix} Q_\tau(X, Z) P_d^{-1'} \right\}^{-1} S \\ &= Y' Q_\tau(X_d, Z_d) \left\{ \begin{pmatrix} X'_d \\ Z'_d \end{pmatrix} Q_\tau(X_d, Z_d) \right\}^{-1} S \\ &= Y' Q_d X_d (X'_d Q_d X_d)^{-1}. \end{aligned}$$

Proof of Lemma 2

Define $B_1 = B_\perp G$ [cf. (A6)], and let B_2 be $n \times (n - r - s)$ matrix of rank $n - r - s$ such that $B_2'(B, B_1) = 0$. Then, by Theorem 3 of Johansen (1992):

(a) $B'\Delta\eta_t$, $B_1'\Delta\eta_t$, and $\bar{A}'\Pi_1\bar{B}_2B_2'\Delta\eta_t + B'\eta_{t-1}$ are $I(0)$,

(b) $((B_2'\Delta\eta_t)', (B_1'\eta_t)')$ is $I(1)$ with no cointegration,

(c) $B_2'\eta_t$ is $I(2)$ with no cointegration,

where $\bar{B}_2 = B_2(B_2'B_2)^{-1}$ and $\bar{A} = A(A'A)^{-1}$. (See also footnote 9 of the present paper.) Hence, we may define

$$C_0 = \begin{pmatrix} B & B_1 & B_2\bar{B}_2'\Pi_1\bar{A} \\ 0 & 0 & B \end{pmatrix},$$

$$C_1 = \begin{pmatrix} B_2 & 0 \\ 0 & B_1 \end{pmatrix},$$

$$C_2 = \begin{pmatrix} 0 \\ B_2 \end{pmatrix}.$$

Then, we can write (17) as

$$\begin{aligned} \Delta^2\eta_t = & J_1^*\Delta^2\eta_{t-1} + \dots + J_k^*\Delta^2\eta_{t-k} + \Pi_1\bar{B}\bar{B}'\Delta\eta_{t-k-1} + \Pi_1\bar{B}_1B_1'\Delta\eta_{t-k-1} \\ & + A(\bar{A}'\Pi_1\bar{B}_2B_2'\Delta\eta_{t-k-1} + B'\eta_{t-k-2}) + \varepsilon_t, \end{aligned}$$

where $\bar{B}_1 = B_1(B_1'B_1)^{-1}$ and where we have used $\bar{B}\bar{B}' + \bar{B}_1B_1' + \bar{B}_2B_2' = I_n$, $A\bar{A}' + A_\perp\bar{A}_\perp' = I_n$ and

$$\begin{aligned} \bar{A}_\perp\Pi_1\bar{B}_2B_2' &= \bar{A}_\perp\Pi_1(\bar{B}\bar{B}' + \bar{B}_\perp B_\perp')\bar{B}_2B_2' \\ &= \bar{A}_\perp\Pi_1\bar{B}_\perp B_\perp'\bar{B}_2B_2' \\ &= FG'B_\perp'\bar{B}_2B_2' \\ &= FB_1'\bar{B}_2B_2' = 0. \end{aligned}$$

Therefore noting that

$$\begin{aligned} w'_{0t} &= (w'_{01t}, (C_0'\tilde{z}_t^{(2)})') \\ &= (\Delta^2\eta'_{t-1}, \dots, \Delta^2\eta'_{t-k}, (B'\Delta\eta_{t-k-1}), (B_1'\Delta\eta_{t-k-1})', \\ &\quad (\bar{A}'\Pi_1\bar{B}_2B_2'\Delta\eta_{t-k-1} + B'\eta_{t-k-2})') \end{aligned}$$

we can write (17) in a stationary VAR(1) representation:

$$w_{0,t+1} = Jw_{0t} + S_1\varepsilon_t,$$

where $S_1 = (I_n, 0)'$ which is an $(nk + r_0) \times n$ matrix, and

$$J = \begin{pmatrix} J_1^* & J_2^* & \dots & J_{k-2}^* & J_{k-1}^* & J_k^* & \Pi_1 \bar{B} & \Pi_1 \bar{B}_1 & A \\ I_n & 0 & \dots & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & I_n & \dots & 0 & 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & I_n & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \dots & 0 & I_n & 0 & 0 & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 & B' & I_r & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 & B'_1 & 0 & I_s & 0 \\ 0 & 0 & \dots & 0 & 0 & \bar{A}' \Pi_1 \bar{B}_2 B'_2 & I_r & 0 & I_r \end{pmatrix}.$$

Note that since w_{0t} is stationary by assumption, all eigenvalues of J are less than unity.

Now, from this VAR(1) representation and (A1), the required convergence results follow by the same argument as that of Theorem 2.2 in Chan and Wei (1988).

Also, write

$$\Sigma_0 = \sum_{h=0}^{\infty} J^h S_1 \Sigma_\varepsilon S_1' J'^h,$$

and the positive definiteness of Σ_0 is proved in the same way as Lemma 5.5.5 of Anderson (1971).

Proof of Lemma 3

All of the convergence results follow from Lemma 2 above and Lemma 2.1 of Park and Phillips (1989) in an entirely analogous way as Lemma 1 and Lemma 2 of Toda and Phillips (1993a).

The nonsingularity of Ω_1 and Ω_2 easily follows from (2.9)–(2.11) of Johansen (1992).

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